

HYPERLIQUID <> LIGHTER PERMISSIVE MID-PRICE GAP REPORT

Report Generated: 2026-04-25 21:57:14

Repository: C:\Users\user\Downloads\Hyper-LighterARB

Data Range: 2026-04-23 08:17:56+00:00 to 2026-04-25 13:55:30+00:00

Fresh Seconds: 114,004

TEST ASSUMPTIONS

Entry: enter whenever mid gap is non-zero.

Direction: short expensive venue, long cheap venue.

Exit: close when |gap| <= 0.25 bps or the spread crosses zero.

Max hold: none. Open positions are marked at end of sample.

Pricing: ideal mid-price fills. Fees, slippage and funding ignored.

Base sizing: \$10,000 per leg.

Scaling test: add \$10,000 per leg whenever gap widens by 1.0 bps, capped at 5 layers.

SINGLE-ENTRY RESULTS

Trades: 4,288

Total PnL: \$5,773.50

Realized PnL: \$5,772.00

End Mark PnL: \$1.50

Win Rate: 83.98%

Average Hold: 123.37s

Profitable Average Hold: 140.84s

WIDENING-SCALE RESULTS

Trades: 4,288

Total PnL: \$13,839.07

Realized PnL: \$13,832.27

End Mark PnL: \$6.80

Win Rate: 84.68%

Average Hold: 123.37s

Average Layers: 1.43

Max Layers Used: 5

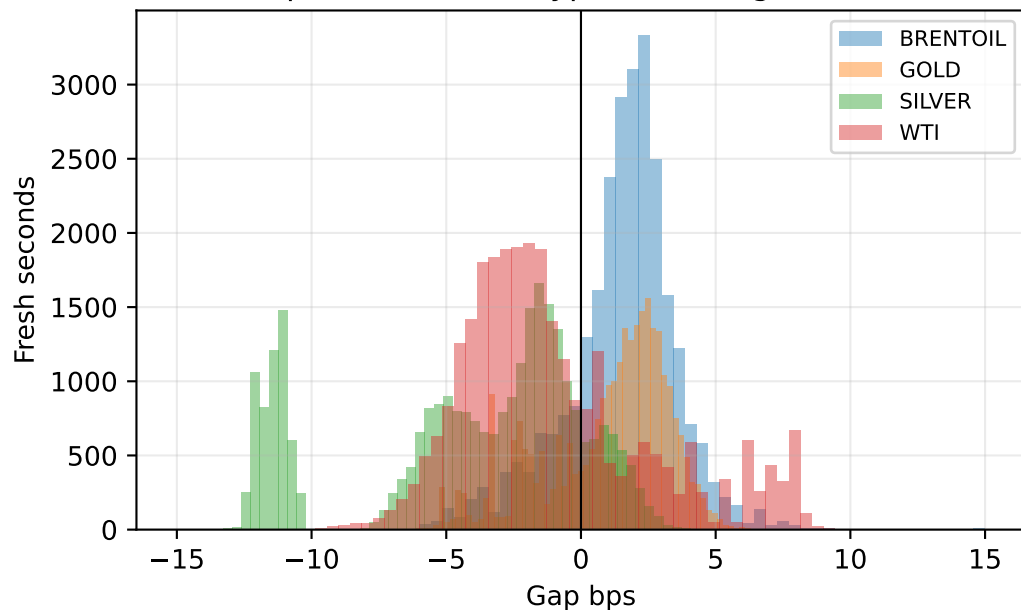
BEST SCALING SENSITIVITY CASE

Step: 0.5 bps, Max Layers: 10

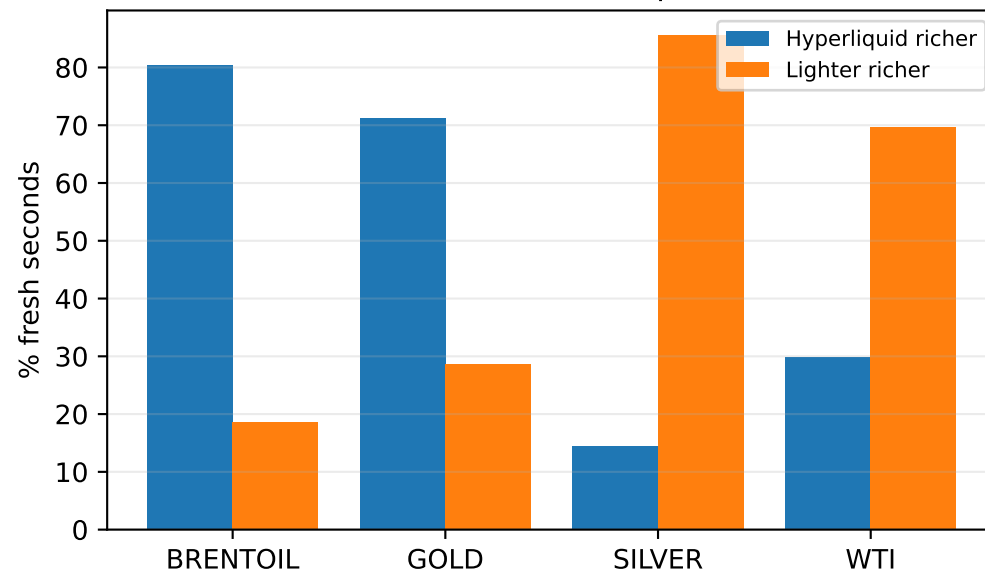
PnL: \$17,277.07, Win Rate: 84.93%, Avg Layers: 1.69

Price Gap and Richness Analysis

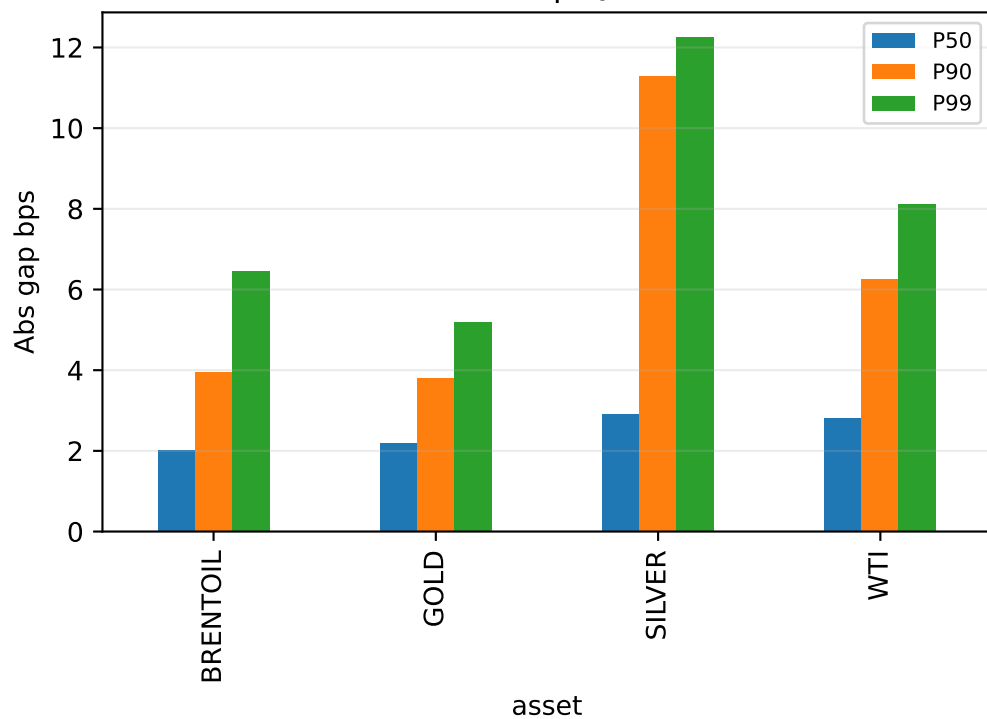
Gap Distribution (Hyper mid - Lighter mid)



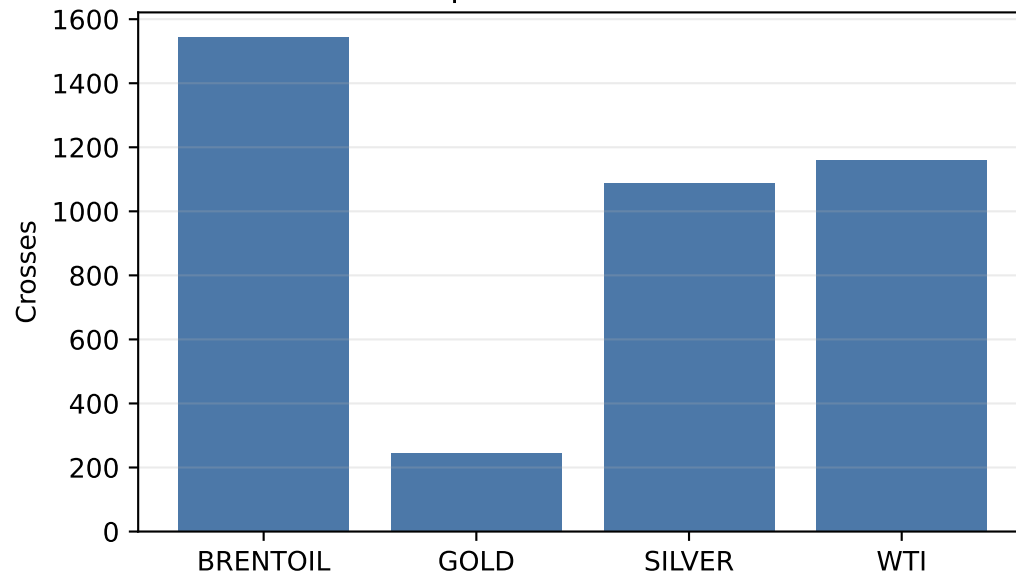
Which Venue Was Expensive



Absolute Gap Quantiles

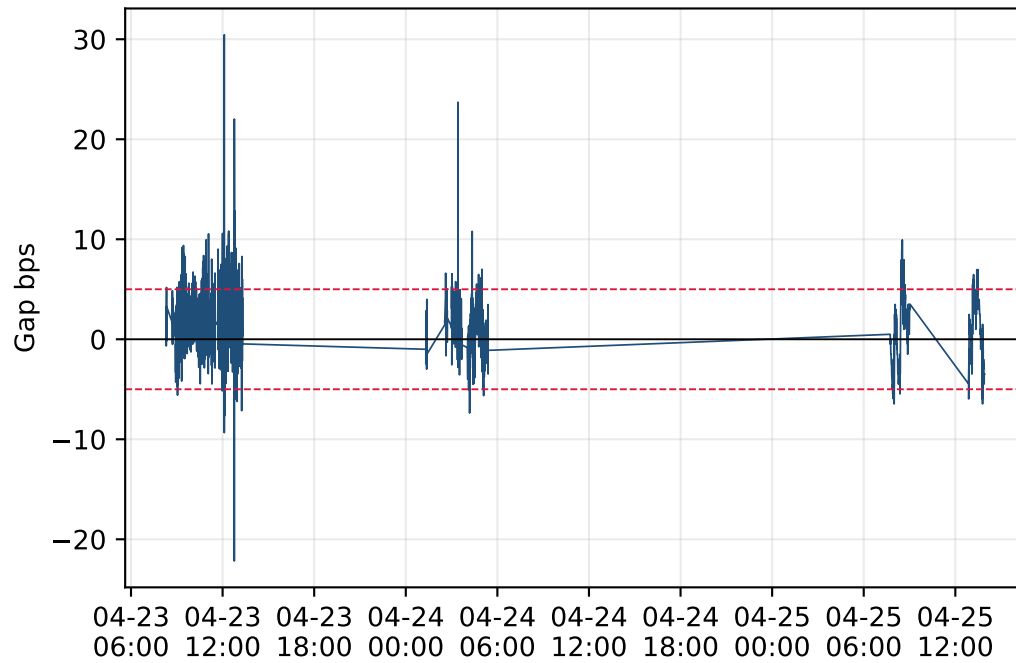


Gap Zero-Cross Count

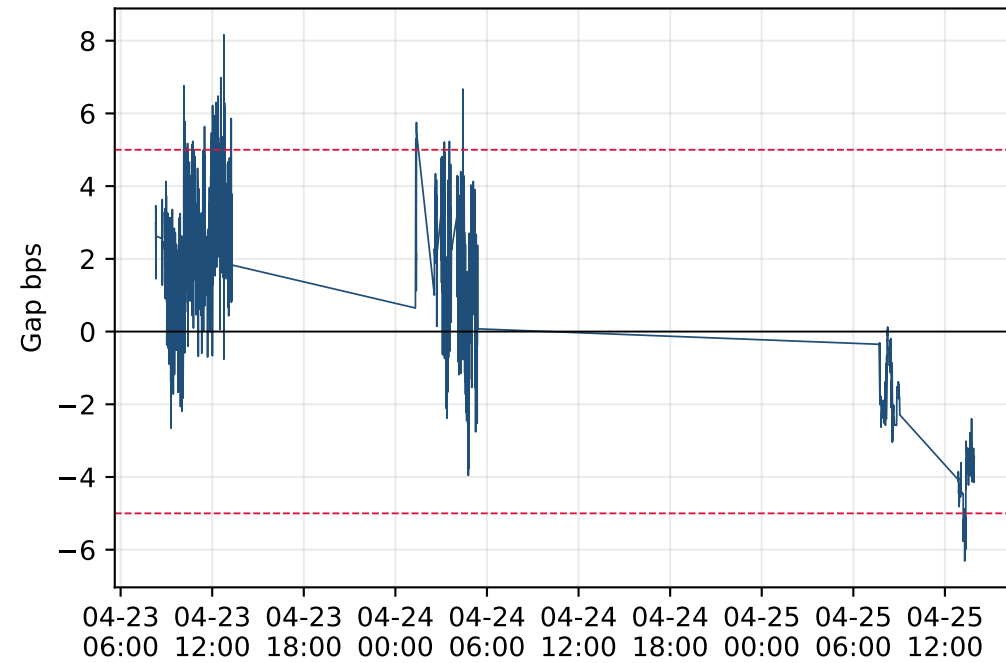


Gap Timing and Clustering

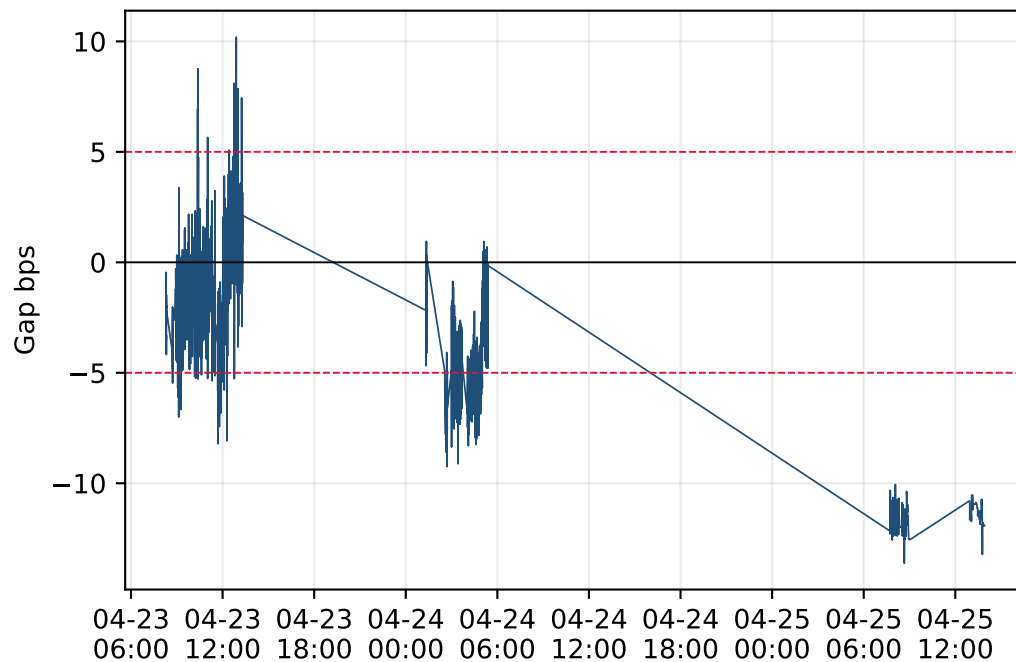
BRENTOIL



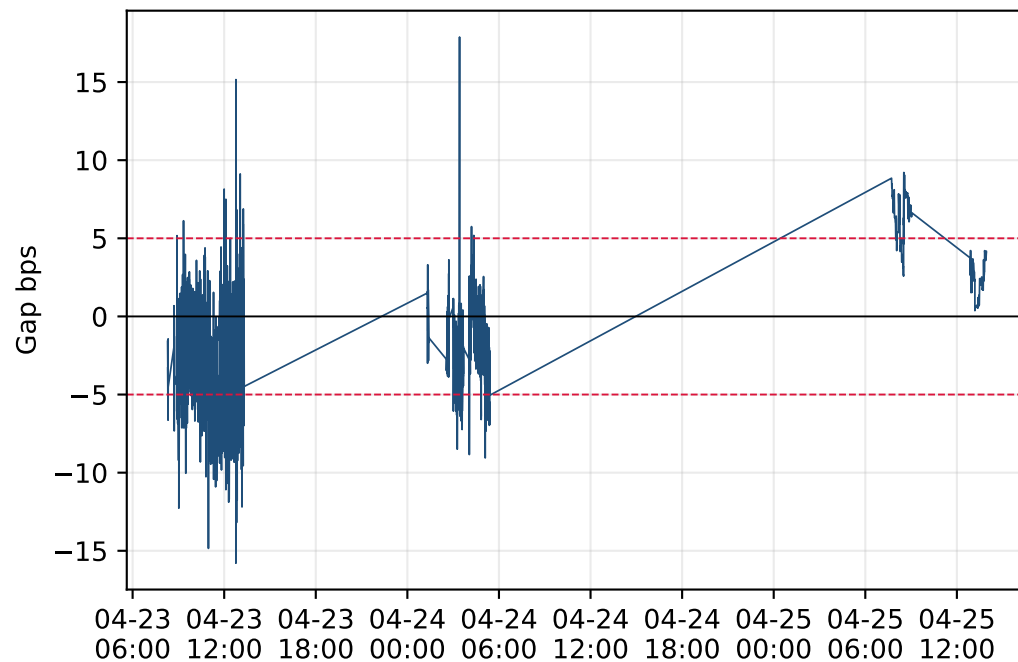
GOLD



SILVER

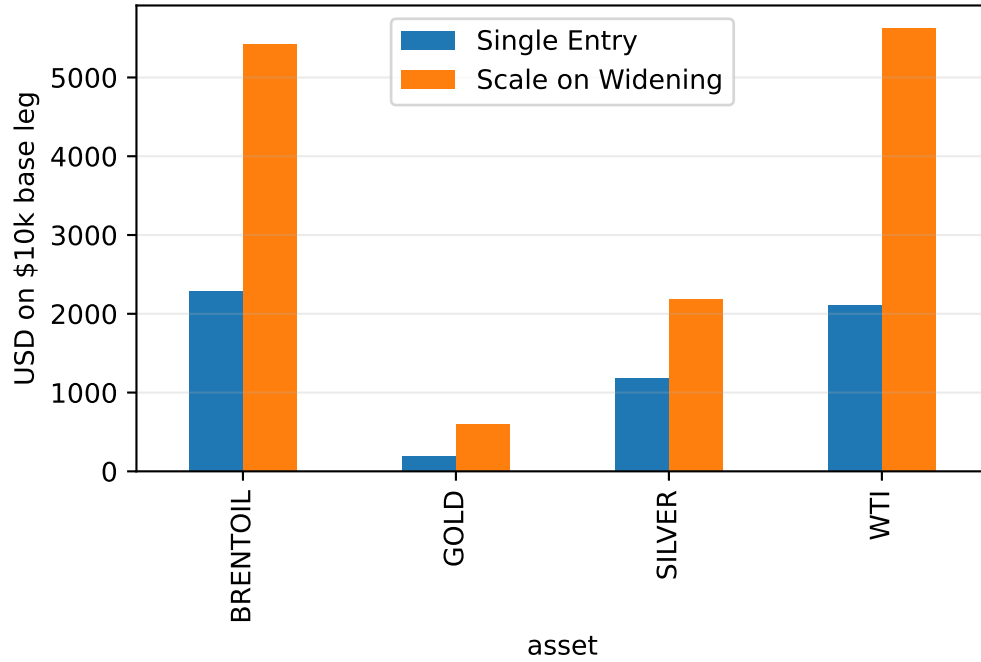


WTI

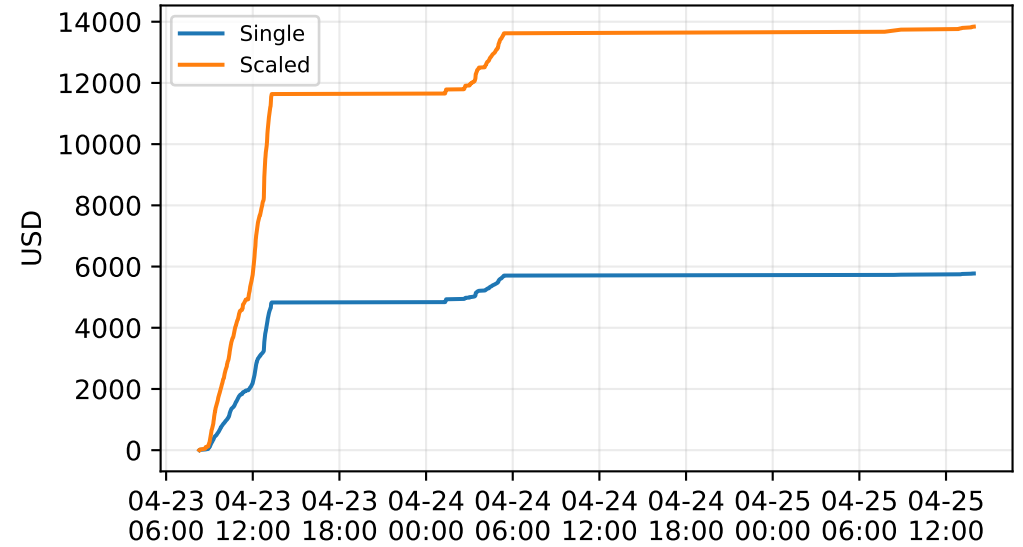


No-Max-Hold Mid-Price Basis Trade Results

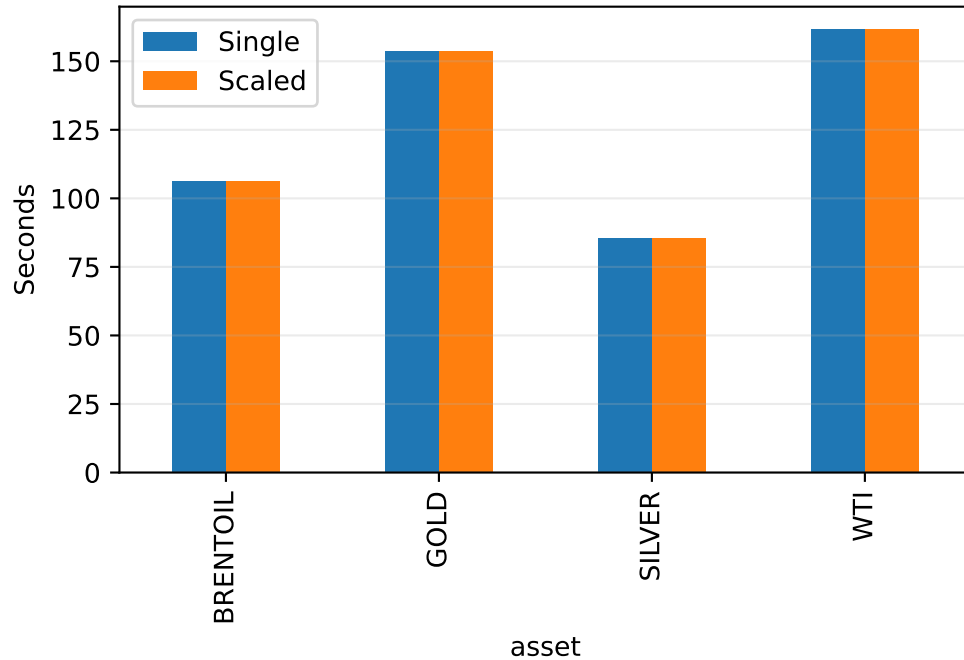
PnL by Asset



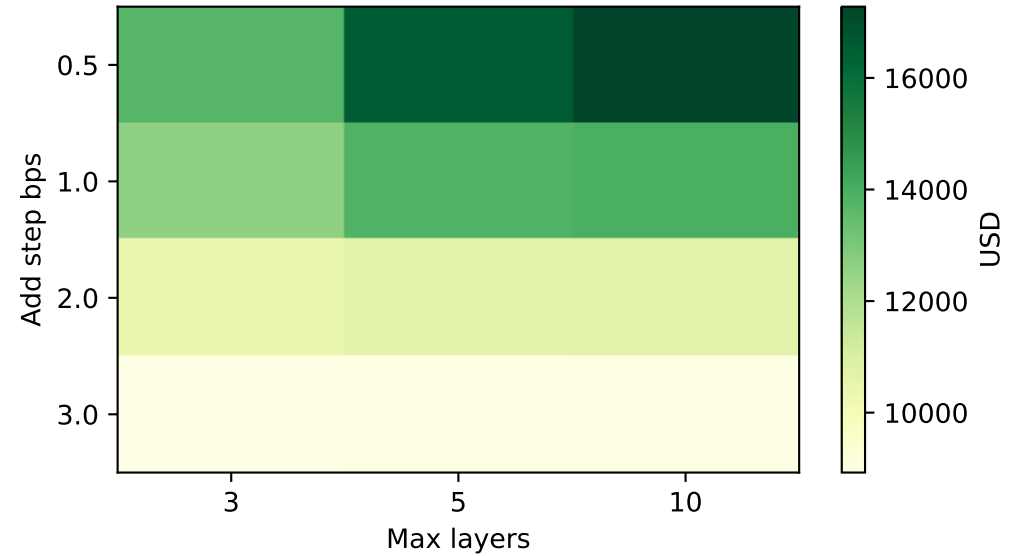
Cumulative PnL



Average Holding Period



Scaling Sensitivity: Total PnL



STRATEGY READOUT

SINGLE-ENTRY BY ASSET

BRENT0IL trades=1391, pnl=\$ 2290.42, win= 91.2%, avg_hold= 106.5s.
GOLD trades= 635, pnl=\$ 193.07, win= 52.1%, avg_hold= 153.7s.
SILVER trades=1083, pnl=\$ 1177.57, win= 92.7%, avg_hold= 85.5s.
WTI trades=1179, pnl=\$ 2112.44, win= 84.6%, avg_hold= 161.8s.

WIDENING-SCALE BY ASSET

BRENT0IL trades=1391, pnl=\$ 5416.59, win= 91.7%, avg_layers=1.49, avg_notional=\$ 14903.
GOLD trades= 635, pnl=\$ 602.63, win= 52.8%, avg_layers=1.22, avg_notional=\$ 12220.
SILVER trades=1083, pnl=\$ 2187.65, win= 94.1%, avg_layers=1.31, avg_notional=\$ 13056.
WTI trades=1179, pnl=\$ 5632.20, win= 85.0%, avg_layers=1.57, avg_notional=\$ 15700.

KEY OBSERVATIONS

1. Removing max hold barely changes the single-entry result because most gaps converge quickly.
2. The average single-entry holding period is about 105 seconds with only four end-marked positions.
3. Scaling into widening improves raw mid-price PnL because large gaps usually mean-revert in this sample.
4. BRENT0IL and WTI contribute most of the PnL; GOLD is weak under this rule.
5. This remains an idealized alpha estimate, not executable PnL, because mid fills ignore spread and queue position.

IMPLEMENTATION IMPLICATION

Use this as the alpha layer: predict/identify spread widening and mean reversion.

Execution should still be maker-biased or selectively aggressive, because crossing every tiny gap was not profitable.